

## DATA INTEGRITY

Data integrity, an area of increasing priority in today's laboratory world and *"The focus area"* when it comes to regulatory compliance such as 21 CFR Part 11 and Eudralex Annex 11. Considering the number of warning letters from USFDA and other such regulatory bodies issued to pharmaceutical and medical devices manufacturers across the globe, Data integrity is the *"Buzz word"* in today's audit.

### WHAT IS DATA INTEGRITY?

Data integrity refers to the overall completeness, accuracy and consistency of data during its entire life cycle. Though seems simple, the whole process of genuinely generating, maintaining & transforming data with completeness and accuracy is a challenging task for any organization.

This white paper discusses the issues faced by organizations in achieving data integrity and briefly describes how and what technology can be effectively used to achieve better data integrity.

### PROBLEMS FACED BY ORGANISATIONS IN UNDERSTANDING DATA INTEGRITY

Most organisations figure out the process of data management which will include capture and archival but fail to understand the core requirement in accomplishing its integrity. For data integrity, it's important to know the "data architecture" of the system. This understanding is key in setting up the focus area and methodologies to achieve better integrity.

### DATA INTEGRITY GUIDELINES BY REGULATORY BODIES

FDA has been adopting the **ALCOA** attributes to verify data integrity

- **Attributable** - Defining Source data and who performed an action on it.
- **Legible** - Permanent recording of information and Access to easy reading any time
- **Contemporaneous**- Recording the date & time when work is performed
- **Original** - Justifying if the information / data is a true copy
- **Accurate** - Is the data accurate, with no errors or editing

To achieve Data Integrity in context to ALCOA as defined above, let us first understand the stages in a data life cycle.

Additional terms based on European Medicines Agency's concept the following acronyms also emerged (CCEA)

- **Complete** - All data including repeat or re-analysis performed on the sample
- **Consistent** - Consistent application of date & time stamps in the expected sequence
- **Enduring** - Recorded in controlled worksheets, laboratory notebooks or electronic media
- **Available** - Available /accessible for review/ audit for the life time of the record

## DATA LIFE CYCLE

A typical data life cycle can be broken up into the following stages:

- ✓ Generation & Recording (including raw data)
- ✓ Processing (including transformation or migration)
- ✓ Retention
- ✓ Retrieval/ Archival
- ✓ Destruction

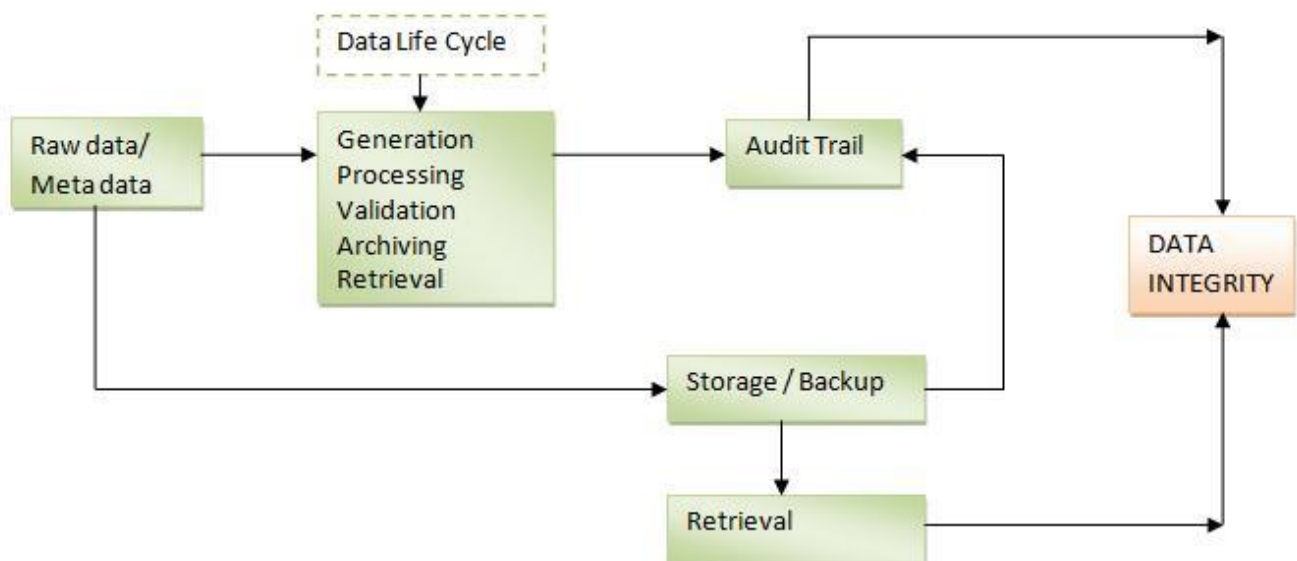


Fig. 1 Data Life Cycle

## ALCOA-CCE TO ACHIEVE FDA & EUDRALEX COMPLIANCE

Adherence to ALCOA-CCE will definitely lead to 21 CFR Part 11 and Eudralex Annex 11 compliance for regulated laboratories. Consider implementing an Electronic Lab Notebook (ELN) and/or a Scientific Data Management System (SDMS) to achieve compliance. Following sections gives you an idea about how these solution help in meeting the guidelines.

### ATTRIBUTABLE: DEFINING SOURCE DATA AND WHO PERFORMED AN ACTION ON IT.

ELN and SDMS systems are instrument & user centric and hence will help easily identifying the data source i.e. which instrument generated the data. Whereas the ELN which is user interface oriented will be able to also track who performed the action. i.e. who generated and collected the data from the instrument.

### LEGIBLE: PERMANENT RECORDING OF INFORMATION AND ACCESS TO EASY READING ANY TIME

ELN and SDMS both systems record in a central database which is a permanent record. Also the database will reside in a controlled server environment. Data will always be available for accessing and reading for authorised persons. Even archived data can be restored and read through these electronic systems in a seamless manner.

Electronic data captured from simple instruments like balance, titrator, pH meter etc. can be easily read even after a very long period of time. This is due to the fact that the communication is direct between the instrument and the software and it is in ASCII. Also data is stored in human readable ASCII format. The life of such data can be eternal (assuming still computers use ASCII).

Whereas the output data generated by complex instruments with complex software as described by [MHRA's data integrity guidance](#) (HPLC, LC-MS) is subject to more variables i.e. meta-data along with primary data is required to make sense on the data. This complexity will lead to trouble in terms of longevity of the data in terms of readability. Systems like ELN and SDMS can capture the data and meta- data in an as-is-basis or original format and also a human readable format (pdf). So it becomes important to maintain a copy of the original software that generated the data during the life time of the data. These systems can maintain such data for a very long period of time

### CONTEMPORANEOUS: RECORDING THE DATE & TIME WHEN WORK IS PERFORMED

Both ELN and SDMS can allow usage or recording of data in a contemporaneous manner i.e. recording of details can happen as and when an activity is being conducted. At the same time they can record the date and time at which the data was captured which is real-time for simple instruments. Whereas for the complex pc controlled instruments the date/time of a file generated by

the instrument becomes the contemporaneous time. SDMS has a real-time file capture mechanism which can capture almost just after the file was generated.

## ORIGINAL: JUSTIFYING IF THE INFORMATION / DATA IS A TRUE COPY

Data captured is always true and directly from instrument (if it is via a port like RS232 or TCP/IP). Also for the complex instrument it is the original file generated by the instrument software that is captured. After a file is generated and if it is modified by some means (mostly through the instrument software) these software systems will capture the new modified file and version it automatically.

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### *CFR Gateway*

*Agaram has developed a unique solution called as “CFR Gateway” to cater to this specific requirement. i.e. “Original” data/meta-data should never be obscured. CFR gateway when used **along with SDMS/ELN** will never allow the instrument software user to delete, rename, save-as, copy, paste files which are monitored by it. This means “Original” data cannot be compromised at any point and time.*

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## ACCURATE: IS THE DATA ACCURATE, WITH NO ERRORS OR EDITING

Data once generated is always pushed to the server. Whereas the local copy/version of the data in the instrument pc can be still used by the instrument user for better convenience. Even if you open an existing data using the original software and edit the data/meta-data, the system will understand that the data has changed and will push the new version to the server. The system has cross check mechanisms such as check-sum calculations to validate the accuracy of the original data versus the data pushed to the server. If there are any such errors the system has failsafe mechanism to retransmit the data to the server.

## COMPLETE: ALL DATA INCLUDING REPEAT OR RE-ANALYSIS PERFORMED ON THE SAMPLE

SDMS: The system versions the raw and meta data files whenever there is a change in the file or whenever a new set of data is received via the RS232 or TCP/IP port. New versions are stored a separate data files with version number.

ELN: Data versioning is available in ELN. i.e. for example if you captured the weight of a sample once and are interested in replacing this weight with another one due to some reason the system will allow you to capture the new weight but will version this new weight and audit trail it with a reason. So any repeat or re-analysis data is always versioned at data level within the ELN

## CONSISTENT: CONSISTENT APPLICATION OF DATE & TIME STAMPS IN THE EXPECTED SEQUENCE

All data capture events and their sequence of capture is data and time stamped. E.g. is possible to go through a pre-determined sequence of events which will be aligned with the actual method of analysis. Each step or

event of recording of data either from instrument or manually recorded in the electronic template is date and time stamped.

## ENDURING: RECORDED IN CONTROLLED WORKSHEETS, LABORATORY NOTEBOOKS OR ELECTRONIC MEDIA

Records are stored in electronic media when using SDMS or an ELN. The ELN provides an electronic equivalent of paper worksheets or notebook to capture data in a controlled manner.

## AVAILABLE: AVAILABLE /ACCESSIBLE FOR REVIEW/ AUDIT FOR THE LIFE TIME OF THE RECORD

Data can always be retrieved even after archival and will be accessible for review or audit for the life time of the record.

## CONCLUSION

Scientific Data Management System (SDMS) and Electronic Lab Notebook (ELN) are key solutions that can help laboratories to achieve better data integrity.

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### *References*

[\*Good Manufacturing Practice \(GMP\) data integrity MHRA Regulations\*](#)

[\*Data integrity definitions and guidance\*](#)

[\*MHRA GMP Data Integrity guidance for the industry 2015\*](#)

[\*FDA Focus on Data Integrity\*](#)

[\*Data Integrity Issues in Pharmaceutical Companies\*](#)

[\*FDA Warning letters on data integrity\*](#)

[\*https://www.parexel.com/files/2614/2184/8648/Schmitt\\_Regulatory\\_Handbook\\_final\\_Jan\\_2015.pdf\*](https://www.parexel.com/files/2614/2184/8648/Schmitt_Regulatory_Handbook_final_Jan_2015.pdf)

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